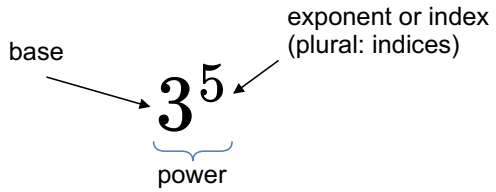
1.01 Indices

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1: Index Laws

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$a^m \times a^n = a^{m+n}$
$a^m \div a^n = a^{m-n}$
$(a^m)^n = a^{mn}$
$(ab)^n = a^n b^n$

Some also refer to the 5 as the ‘power’. But this would lead to unfortunate ambiguous phrases like “when we multiply two powers together, we add the powers” instead of “when we multiply two powers together, we add the indices”.

Simplify $(a^3)^2 \times 2a^2$

Simplify $(4x^3y)^3$

Simplify $2x^2(3+5x) - x(4-x^2)$

Simplify $\frac{x^3-2x}{3x^2}$

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Test Your Understanding

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1 Simplify $\left(\frac{2a^5}{a^2}\right)^2 \times 3a$

2 Simplify $\frac{2x+x^5}{4x^3}$

3 Expand and simplify
$2x(3-x^2) - 4x^3(3-x)$

4 Simplify $2^x \times 3^x$

1.01 Indices

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Exercise 1A – Pearson Pure Y1 page 4

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2 Expand and simplify if possible:

o $x(3x^2 - 2x + 5)$ p $7y^2(2 - 5y + 3y^2)$ q $-2y^2(5 - 7y + 3y^2)$

r $7(x - 2) + 3(x + 4) - 6(x - 2)$ s $5x - 3(4 - 2x) + 6$

t $3x^2 - x(3 - 4x) + 7$ u $4x(x + 3) - 2x(3x - 7)$ v $3x^2(2x + 1) - 5x^2(3x - 4)$

3 Simplify these fractions:

d $\frac{8x^3+5x}{2x}$ **e** $\frac{7x^7+5x^2}{5x}$ **f** $\frac{9x^5-5x^3}{3x}$

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Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 [MAT 2006 1A] Which of the following numbers is largest?

$((2^3)^2)^3$, $(2^3)^{(2^3)}$, $2^{((3^2)^3)}$, $2^{(3^{(2^3)})}$

2 [MAT 2006 1A] Let $N = 2^k \times 4^m \times 8^n$ where k, m, n are positive whole numbers. Then N will definitely be a square number whenever:

k is even;
$k + n$ is odd;
k is odd but $m + n$ is even;
$k + n$ is even.

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4: Negative and Fractional Indices

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$$a^0 = 1$$
$$a^{\frac{1}{m}} = \sqrt[m]{a}$$
$$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$$
$$a^{-m} = \frac{1}{a^m}$$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.
Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $27^{-\frac{1}{3}}$

Evaluate $32^{\frac{2}{5}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

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Writing a surd using indices

18

If $9\sqrt{3} = 3^k$, determine the value of k.

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Further Examples

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[Edexcel IGCSE May14-4H]
Given that

$$\left(2^{\frac{1}{2}}\right)^{n} = \frac{2^{x}}{8^{y}}$$

Express n in terms of x and y.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^{x}$, express the following in terms of y.

$$\frac{1}{4^{2x-3}}$$

Write your expression in its simplest form.

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Test Your Understanding

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Edexcel Paper 2 – May 2019

Given $2^{x} \times 4^{y} = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

1.01 Indices

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Exercise 1D

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4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants. **a** $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

[MAT 2007 1A] Let r and s be integers. Then $\frac{6^{r+s}\times 12^{r-s}}{8^r \times 9^{r+2s}}$ is an integer if $r+s \leq 0$; $s \leq 0$; $r \leq 0$; $r \geq s$.